San Diego Heat Risk Explorer User Guide

A primer on variables, methods, and map interpretation.

The Heat Risk Explorer is an interactive heat-risk map of San Diego County, computed at the census-tract level. Select the variables you are interested in (heat exposure, demographics, health conditions, air conditioning access, green space proximity, income, and others) and the tool returns a composite heat risk score and the dominant driver for each census tract.

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What goes into the score

The score combines three components: **heat exposure** (the physical heat experienced in a given census tract), **population sensitivity** (who lives there and how susceptible they are), and **adaptive capacity** (the ability of residents to cope with heat stress). Each component draws on multiple variables, selected by the user of the tool.

Heat Exposure

- **Daytime highs:** mean daily maximum, 85th and 99th percentile of daily maximums (the hottest 15% and 1% of days), annual maximum, and Landsat-derived land surface temperature.
- **Heat stress:** 85th and 99th percentile Wet-Bulb Globe Temperature, which combines temperature, humidity, sun, and wind into a single heat-safety index.
- **Nighttime lows:** mean overnight minimum, 90th-percentile warm nights, and the share of nights without cooling relief. Warm nights are associated with elevated heat-illness risk, especially in older adults.
- **Humidity:** mean relative humidity, mean humidity during hot hours, and 95th-percentile dewpoint. High humidity reduces evaporative cooling and raises apparent temperature.

Population Sensitivity

- **Age:** share of residents under 18 and over 65.
- **Chronic conditions:** heart disease, asthma, COPD, diabetes, hypertension, depression, cognitive difficulty, and other heat-aggravated conditions (CDC PLACES Data).
- **Social vulnerability (Social Vulnerability Index):** limited English, disability, poverty, no vehicle, housing-cost burden, single-parent households, no high-school diploma, uninsured, and additional demographic indicators.
- **Air quality and life expectancy:** ozone, PM2.5, diesel particulates, and tract-level life-expectancy percentile.

Variables listed above are examples; additional CDC PLACES Data and Social Vulnerability Index variables can be toggled individually.

Adaptive capacity

- **Air conditioning:** share of homes with any AC, with central AC, or with no AC.
- **Built environment:** tree canopy, park access, green space, housing age (pre-1980 and pre-1960 share), and renter share.
- **Resources and services:** income, education, health insurance, healthcare access, food access (supermarkets and retail), commute time, voter participation, and other Healthy Places Index domains.

Higher adaptive capacity lowers the composite score. The categories above show representative variables; further Healthy Places Index domains are available in the variable picker.

How the components combine

Each pillar score is a tract's percentile rank within San Diego County for a weighted combination of selected variables. The composite is the mean of the three pillars (heat exposure, population sensitivity, and adapative capacity), producing a single value per tract between 0–100.

Correlated variables are not double-counted. Lower-income tracts tend to have older housing, less tree cover, and higher chronic-illness rates. A direct sum would weight that shared pattern multiple times. The tool uses Principal Component Analysis (PCA) to identify the underlying patterns and weight each one once.

Adaptive-capacity variables enter the score with the opposite sign. A tract with strong protective resources receives a lower composite score even when its exposure is high.

Reading the map

Colors

Darker color means higher risk. The composite map uses red (highest = darkest red). Single-pillar views use the pillar's color: red for heat exposure, yellow for population sensitivity, teal for adaptive capacity.

Score interpretation

Scores are *relative percentiles*, not absolute risk. A score of 75 means the tract is at higher risk than 75% of San Diego County tracts. It does not mean 75% of residents will be harmed.

Map layers

- **Composite Risk:** the 0–100 score combining all three pillars (default view).
- **Dominant Driver:** tracts are colored by which pillar contributes most to the score. This distinguishes hotspots driven by heat exposure from those driven by population sensitivity or low adaptive capacity. Click on a census tract to see the scores for each of the three pillars.
- **Primary Spatial Patterns:**This drop down menu shows the independent patterns contained within the data. Hover over the blue info icons to see which variables contribute to each pattern.

Tract panel

Click a tract for its three pillar scores, its composite score, and its dominant driver.

Future scenarios

Three buttons at the top of the Climate Scenario panel switch the temperature variables between time periods:

- **Present Day:** observed weather-station data.
- **2050:** mid-century mean (2035–2064) from LOCA2-downscaled CMIP6 projections.
- **2080:** end-of-century mean (2065–2094) from LOCA2-downscaled CMIP6 projections.

Selecting 2050 or 2080 expands an emissions-pathway picker with three Shared Socioeconomic Pathways (SSPs; see Carbon Brief explainer):

- **Moderate Emissions (SSP 2-4.5):** emissions decline but not aggressively.
- **High Emissions (SSP 3-7.0):** emissions remain high.
- **Very High Emissions (SSP 5-8.5):** unchecked fossil-fuel use.

Compare Scenarios Side-by-Side (button below the picker) opens a paired view: the composite-risk map under one scenario on the left, and the change in heat-exposure pillar rank between two chosen scenarios on the right (B minus A). The two scenarios on either side can be set independently.

Population sensitivity and adaptive-capacity variables are held at present-day values; the tool does not project future changes to these variables.

Examples

Each example narrows one pillar to a single question, then leaves the other two pillars alone (all variables remain on by default) so the composite still reflects who is hot, who is vulnerable, and who can cope.

1. Where are children most at risk due to heat stress?

1. Click the **Population Sensitivity** header checkbox to clear that pillar, then re-check only **Age 17 & Under** (under *Demographics*).
2. Select all variables for Heat Exposure and Adaptive Capacity.
3. Click **Build Composite Risk Map**.
4. The Population Sensitivity score is now driven entirely by the population under 17. Census tracts that are high risk (darker) **and** driven primarily by population sensitivty are places where children are most at risk due to heat stress.

2. Where would people benefit most from subsidies to purchase AC units?

1. Click the **Adaptive Capacity** header checkbox to clear that pillar, then re-check only **AC Prevalence** (under *Air Conditioning*).
2. Select all variables for Heat Exposure and Population Sensitivity.
3. Click **Build Composite Risk Map**.
4. The Adaptive Capacity score is now a pure ranking of how AC-poor each tract is. The darkest red tracts are priority areas for an AC-subsidy program: hot, sensitive, and currently lacking AC.

3. Where is nighttime heat most dangerous?

1. Click the **Heat Exposure** header checkbox to clear that pillar, then re-check the Nighttime Heat variables: **Mean Overnight Minimum Temperature**, **90th Percentile Overnight Minimum Temperature**, and **No-Relief Night Fraction**.
2. Leave Population Sensitivity and Adaptive Capacity alone.
3. Click **Build Composite Risk Map**.
4. The Heat Exposure score is now a pure ranking of how warm a tract's nights are. The darkest red tracts have severe overnight heat alongside vulnerable populations and limited capacity to cool down.

To report a bug or request a feature, open an issue here.

Data sources

VARIABLE	SOURCE
Weather observations	MesoWest / Synoptic Data
Future temperature projections	LOCA2 downscaled CMIP6
Land surface temperature	USGS Landsat
Chronic disease prevalence	CDC PLACES
Social vulnerability	CDC SVI 2022
Income, education, healthcare, housing	Healthy Places Index (HPI)
Demographics and housing	ACS 5-year estimates
Air conditioning prevalence	Romitti et al.; Ahn et al. (modeled)

Each variable name in the tool has a small `src` link that opens the original source.

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